

Python Practice Tasks

For Loops, List Comprehension, Nested Loops, Break/Continue/Pass,
Enumerate, While Loops & Try/Except

Session [3]

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18 Questions — Solved

Section 1 — For Loop & List Comprehension

Q1. Classify Numbers

Given `lst = [12, 7, 23, 4, 18, 9, 31, 6, 15, 2]`: print every number greater than 10. Build a list "big" containing numbers greater than 10, and a list "small" containing numbers 10 or less.

Answer:

```
lst = [12, 7, 23, 4, 18, 9, 31, 6, 15, 2]
for i in lst:
    if i > 10:
        print(i)

big = [i for i in lst if i > 10]
small = [i for i in lst if i <= 10]
print(big, small, sep="\n")
```

Q2. Multiplication Table using range()

Using `range()` (no manual list), print the multiplication table of a number (`num = 7`) from 1 to 10.

Answer:

```
num = 7
for i in range(1, 11):
    print(f"{i}X{num} = {i*num}")
```

Q3. Sum & Count of Even Numbers

Given `lst = [3, 8, 15, 22, 6, 11, 40, 9]`: compute the sum of all numbers using a for loop (without `sum()`), and count how many are even using a counter.

Answer:

```
lst = [3, 8, 15, 22, 6, 11, 40, 9]

total = 0
for i in lst:
    total += i
print(total)

counter = 0
for i in lst:
    if i % 2 == 0:
        counter += 1
print(counter)
```

Q4. Clean Text from Digits

Given text = "H3ll0 W0rld! Th1s #is \$a T3st%" and digits = "0123456789": remove all digits using a plain for loop, then repeat using a list comprehension with ".join(...)".

Answer:

```
text = "H3ll0 W0rld! Th1s #is $a T3st%"
digits = "0123456789"

no_digit = ''
for i in text:
    if i not in digits:
        no_digit += i
print(no_digit)

no_digit2 = [i for i in text if i not in digits]
print(''.join(no_digit2))
```

Q5. Filter Names by Length & Capitalize

Given names = ["Ali", "Sara", "Mo", "Yara", "Ahmed", "Lina", "Omar"]: build a list comprehension of names with more than 3 letters, and another with every name in uppercase.

Answer:

```
names = ["Ali", "Sara", "Mo", "Yara", "Ahmed", "Lina", "Omar"]

moreThreeNames = [i for i in names if len(i) > 3]
print(moreThreeNames)

upperNames = [i.upper() for i in names]
print(upperNames)
```

Q6. Count & Extract Vowels

Given text = "Egypt is beautiful and Egypt has a great history" and vowels = "aeiouAEIOU": count vowels using a for loop, then build a list comprehension of every vowel found (with repetition).

Answer:

```
text = "Egypt is beautiful and Egypt has a great history"
vowels = "aeiouAEIOU"

counter = 0
for i in text:
    if i in vowels:
        counter += 1
print(counter)

findVowels = [i for i in text if i in vowels]
print(findVowels)
```

Section 2 — Nested For Loops

Q7. Product of Each Row + Total Sum

Given matrix = [[5, 10, 2], [1, 8, 9], [4, 4, 4]]: multiply the elements of each row together (wrap each row's result in [] to keep a matrix-like shape), then sum all the row results into a total.

Answer:

```
matrix = [[5, 10, 2], [1, 8, 9], [4, 4, 4]]

row_multiply = []
total = 0

for i in matrix:
    miniMultiaple = 1
    for j in i:
        miniMultiaple *= j
    row_multiply.append([miniMultiaple])

print(f"row_multiply : {row_multiply}")

for i in row_multiply:
    for j in i:
        total += j

print(f"total = {total}")
```

Q8. Multiply Every Element by a Number

Given matrix = [[1,2,3],[4,5,6],[7,8,9]] and factor = 3: build a new matrix (same shape) where every element is multiplied by factor, using a nested for loop.

Answer:

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]]

factor = 3

result = []

for i in matrix:
    miniResult = []
    for j in i:
        miniResult.append(j * factor)
    result.append(miniResult)

print(f"Result : {result}")
```

Q9. Count & Extract Even Numbers per Row

Given matrix = [[3,8,15],[22,6,11],[40,9,12]]: for each row, count the even numbers and collect them, using a nested for loop.

Answer:

```
matrix = [[3, 8, 15],
          [22, 6, 11],
          [40, 9, 12]]

row_evens_count = []
evens = []

for i in matrix:

    evens_count = 0
    mini_evens = []

    for j in i:
        if j % 2 == 0:
            evens_count += 1
            mini_evens.append(j)

    row_evens_count.append(evens_count)
    evens.append(mini_evens)

print(f"Row Evens Counter : {row_evens_count}\n"
      f"Evens : {evens}")
```

Section 3 — Break / Continue / Pass

Q10. Skip Negative Numbers (continue)

Given lst = [5, -3, 8, -1, 12, -7, 4, 9]: collect only the positive numbers, skipping negatives with continue.

Answer:

```
lst = [5, -3, 8, -1, 12, -7, 4, 9]
positive_lst = []

for i in lst:
    if i < 0:
        continue
    positive_lst.append(i)

print(positive_lst)
```

Q11. Stop at First Number Divisible by 7 (break)

Given lst = [4, 9, 15, 21, 6, 28, 3]: print each number, and stop the loop immediately (break) once you hit a number divisible by 7.

Answer:

```
lst = [4, 9, 15, 21, 6, 28, 3]

for i in lst:
    print(i)
    if i % 7 == 0:
        break
```

Q12. Placeholder Structure (pass)

Given lst = [1,2,...,10]: for even numbers leave the branch empty for now using pass; for odd numbers, print them.

Answer:

```
lst = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

for i in lst:
    if i % 2 == 0:
        pass
    else:
        print(i)
```

Q13. Combine All Three (pass, continue, break)

Given lst = [3,-5,8,0,-2,15,7,-9,22,6]: skip negatives (continue), leave zero untouched (pass, no print), stop the loop at any number greater than 20 (break), otherwise print the number.

Answer:

```
lst = [3, -5, 8, 0, -2, 15, 7, -9, 22, 6]

for i in lst:
    if i < 0:
        continue
    elif i == 0:
        pass
    elif i > 20:
        break
    else:
        print(i)
```

Section 4 — Enumerate

Q14. Enumerate — Numbering, Filtering & Dict Conversion

Given names = ["Ali", "Sara", "Mo", "Yara", "Ahmed"]: (1) print each name with its number starting from 1 using enumerate. (2) Build a list comprehension of names whose index is even. (3) Convert the enumerate result into a dictionary (index -> name).

Answer:

```
names = ["Ali", "Sara", "Mo", "Yara", "Ahmed"]

for i, v in enumerate(names, 1):
    print(f"{i} => {v}")

even_names = [v for i, v in enumerate(names) if i % 2 == 0]
print(even_names)

print(dict(i for i in enumerate(names)))
```

Section 5 — While Loop

Q15. Number Guessing Game (while + break)

secret = 37. Using while True, repeatedly ask the user for a number. Print whether it is smaller or bigger than the secret; on a correct guess print a success message and break. Finally print how many attempts it took.

Answer:

```
secret = 37
counter = 0

while True:
    try:
        number = int(input("Enter number: "))
    except:
        print("Please enter a valid number only")
        continue

    counter += 1

    if number < secret:
        print("Smaller than the PIN")
    elif number > secret:
        print("Bigger than the PIN")
    else:
        print("Congratulations! You guessed the right number")
        break

print(f"counter = {counter}")
```

Q16. Largest Number Divisible by 3 and 5 (while only)

Ask the user for n. Starting from n-1 and going downward using while only (no for), find the first (largest) number below n divisible by both 3 and 5, print it and stop. If the counter reaches zero without finding one, print that no suitable number exists.

Answer:

```
while True:
    try:
        number = int(input("Please, Enter number : "))
        break
    except:
        print("please enter a valid number only")

i = number

while True:
    i -= 1
    if i % 15 == 0:

        if i == 0:
            print("There is no suitable number")
            break

        print(f"The number that is divisible by 3 and 5 : {i}")
        break
```

Q17. Collect Numbers with While, Process with For

Use an outer while loop to repeatedly collect numbers from the user into a list until they type "done". Then use a for loop to print the even numbers and their total. If no numbers were entered, print a message saying so.

Answer:

```
lst = []

while True:
    try:
        number = input("Enter integer number : ")

        if number.lower() == "done":
            break
        else:
            number = int(number)

    except:
        print("Please, Enter a valid number only : ")
        continue

    lst.append(number)

if len(lst) == 0:
    print("No numbers were entered")
```



```

else:
    print(f"\n1st = {lst}\n")

    total = 0

    for i in lst:
        if i % 2 == 0:
            print(f"even number {i}")
            total += i

    print(f"total of even number = {total}")

```

Section 6 — Try / Except / Else / Finally

Q18. Safe Fruit Calculator (try / except / else / finally)

Given prices = {"apple": 10, "banana": 5, "mango": 20}; ask the user for a fruit name and look up its price (catch `KeyError` if it doesn't exist). If found, ask for a quantity (catch `ValueError` if it's not a number). If both succeed, compute and print the total inside `else`. Always print a closing message inside `finally`.

Answer:

```

prices = {"apple": 10, "banana": 5, "mango": 20}

name_fruits = str(input("name fruit is :"))

try:
    print(f"price {name_fruits.lower()} is {prices[name_fruits.lower()]}")
    quantity = float(input("Enter Quantity : "))

except KeyError:
    print("This fruit doesn't exist")

except ValueError:
    print("Please, enter integer number to this quantity")

else:
    total = prices[name_fruits.lower()] * quantity
    print(f"total price = {total:.2f} L.E")

finally:
    print("The operation is over")

```